

Supporting Data and Code: Reproducibility Documentation

Investment concentration and financing in the AI boom

Companion to the manuscript submission

1 Purpose and scope

This document describes the supporting data-and-code archive that accompanies the manuscript. The archive is a self-contained, **offline**, **deterministic** reproduction: from a single command it regenerates every reported number, table and figure from frozen inputs, with **no network access, no API keys and no paid calls**. The permanent citable version is the Zenodo deposit at [[DOI PENDING]] (code MIT-licensed).

A strict wall separates two phases. An *offline analysis* reproduces every claim from the frozen inputs (documented here). A *collection* phase originally queried public SEC and BLS endpoints and is retained for transparency only (Section 8). Because the SEC's *frames* interface returns the most recently filed value for each fact, re-collecting today would pick up subsequent filer restatements and would not reproduce the frozen inputs; collection is therefore deliberately not part of the reproducible core.

2 Quick start

```
pipinstall-rrequirements.txt
./verify.sh
```

Expected result: the archive verifies against its SHA-256 manifest, the analysis re-runs from the frozen inputs, the regenerated `claims.json` and figure are byte-identical to the committed ones, every number in the manuscript reconciles to `claims.json`, the manuscript is within the journal's length limit, and the script exits 0. Total runtime is under ten seconds on a standard laptop. No network access is required at any point.

3 Archive layout

<code>verify.sh</code>	one-command offline verification (referee path; Section 7)
<code>reproduce.sh</code>	same pipeline plus rebuilding <code>manuscript.pdf</code> (author path)
<code>MANIFEST.sha256</code>	SHA-256 of every archived file, in <code>sha256sum-c</code> format
<code>requirements.txt</code>	<code>matplotlib</code> (figure only); every number needs the standard library alone
<code>environment.txt</code>	exact versions that produced the committed results
<code>manuscript.tex</code> / <code>.pdf</code>	the submission (elsarticle; reproducible source)
<code>manuscript_blinded.tex</code> / <code>.pdf</code>	blinded review copy (identifiers and metadata removed)
<code>COVER_LETTER.md</code> / <code>.tex</code> / <code>.pdf</code>	cover letter
<code>highlights.txt</code>	journal highlights (5 bullets)
<code>figures/Figure_1.pdf</code> / <code>.png</code>	the figure, vector and 200-dpi raster
<code>claims.json</code>	canonical single source of truth: every number the paper asserts

code/	collectors, engine, claims builder, figure builder, gates (Section 5)
data/frozen/	8 frozen source files, read-only (Section 4)
data/SHA256SUMS.txt	SHA-256 of every frozen source file
README.md, SUBMISSION.md, DATA _AVAILABILITY.md	orientation, journal checklist, data statement
CITATION.cff, .zenodo.json, LI CENSE	citation and deposit metadata, licences

4 Data (inputs and metadata)

4.1 Frozen inputs

Every input the pipeline reads is a frozen artifact. The analysis is a pure function of these files; it contains no random component and no seed.

File	Records	Unit of observation / contents
capex_frames.json	88,097	firm-year capital expenditure, 2009–2025, two tags: <code>PaymentsToAcquirePropertyPlantAndEquipment</code> , <code>PaymentsToAcquireProductiveAssets</code>
composition_frames.json	161,802	firm-year operating cash flow, R&D, repurchases, dividends, 2013–2025
revenue_frames.json	95,605	firm-year revenue (three tags) and continuing-operations cash flow, 2013–2025; revenue ranks the universe
financing_frames.json	63,293	firm-year long-term-debt issuance and repayment, and cash paid for acquisitions, 2013–2025
alt_frames.json	270,979	alternate tags for every key series (nine tags), 2013–2025; the union is what makes the measure tag-robust
ai_intensity.json	285	one scored 10-K per firm: dictionary occurrences, total words, AI intensity, filing period
sic_codes.json	500	registrant SIC code and description, as filed
deflator_WPUFD41312.json	13	BLS annual index values, 2013–2025

4.2 Provenance and integrity

`data/SHA256SUMS.txt` fixes every frozen source file; `MANIFEST.sha256` fixes every file in the archive, including the code and the manuscript. The collection scripts write their outputs read-only and refuse to overwrite an existing frozen file, so a silent change to an input is detectable before the analysis runs. `verify.sh` checks both manifests before it runs anything.

4.3 Data sources

All inputs are public and free. Firm-level data were retrieved on 2–4 September 2026 from the SEC’s XBRL *frames* interface, one call per tag and calendar year, at <https://data.sec.gov/api/xbrl/frames/us-gaap/<tag>/USD/CY<year>.json>. Registrant classifications and 10-K

locations come from the submissions API, <https://data.sec.gov/submissions/CIK<10-digit-CIK>.json>, and 10-K documents from EDGAR at <https://www.sec.gov/Archives/edgar/data/<CIK>/<accession>/<document>>; see <https://www.sec.gov/edgar/sec-api-documentation>. The SEC requires only a descriptive `User-Agent` header and no API key. The deflator is Bureau of Labor Statistics series `WPUFD41312`, final demand–private capital equipment, annual averages `[[DEFLATOR RETRIEVAL]]`. No proprietary database is used at any stage.

5 Code (what it consumes and runs)

5.1 Analysis scripts

All code is in `code/`; each module is deterministic and reads only frozen data.

Script	Purpose	Produces
<code>engine.py</code>	measurement engine: unions alternate XBRL tags, deflates to 2025 dollars, aggregates firms; defines the tag lists for every series	imported by the others
<code>make_claims.py</code>	the whole analysis: coverage ladder, placebo windows, sample funnel, sectoral decomposition, leave-one-out, AI-intensity cuts, concentration	<code>claims.json</code>
<code>make_figure.py</code>	the three panels of Figure 1 from the same frozen inputs	<code>figures/Figure_1.pdf</code> , <code>.png</code>
<code>build_gate.py</code>	reads every number out of the manuscript and requires each to be reproducible from <code>claims.json</code> ; anchors headline figures and both table bodies to their source; cross-checks quantities reported twice	pass / fail
<code>wordcount.py</code>	journal length count (main text, table notes, figure captions, references)	pass / fail
<code>make_manifest.py</code>	regenerates <code>MANIFEST.sha256</code> from the tracked file list	<code>MANIFEST.sha256</code>
<code>make_blinded.py</code>	strips identifiers and PDF metadata for the blinded copy	<code>manuscript_blinded.*</code>
<code>reconcile.py</code>	(audit) matches constructed values to as-filed 10-K values for the five focal firms	reconciliation counts
<code>collect_*.py</code>	original data collection (Section 8; not run during reproduction)	<code>data/frozen/*</code>

5.2 Pipeline order

`verify.sh` runs the modules in dependency order; `claims.json` is timestamp-free, so re-runs are byte-identical:

```
make_claims→make_figure→build_gate→wordcount
```

6 Results (generated outputs)

Running the pipeline regenerates the following. In a clean checkout these already exist; the reproduction overwrites them and confirms they are unchanged.

Output	Contents
<code>claims.json</code>	every number the manuscript asserts. Keys used by the paper include <code>ladder</code> (Table 1), <code>placebo_ladder</code> (Table 2), <code>funnel</code> and <code>excluded</code> (the sample account), <code>composition</code> (the sectoral decomposition), <code>firm_level</code> and <code>leave_one_out_cum</code> (Table A3), <code>ai_filter</code> (Table A1 panel B), <code>alt_tag</code> , <code>annual</code> , <code>concentration</code> , <code>hashes</code>
<code>figures/Figure_1.pdf</code> , <code>.png</code>	(a) capital expenditure as a share of operating cash flow for the five largest investors; (b) Herfindahl index of capital expenditure within the 500, with and without those five; (c) coverage of the increment cumulated through each year
<code>manuscript.pdf</code>	the compiled submission (<code>reproduce.sh</code> only)

7 Reproduction and verification

`verify.sh` performs, and exits non-zero if any stage fails:

1. **Integrity.** Every archived file is checked against `MANIFEST.sha256`; a modified frozen input, script or manuscript fails here, before anything runs.
2. **Regeneration.** The analysis runs from `data/frozen/` only.
3. **Byte-identical re-run.** The regenerated `claims.json` and `figures/Figure_1.png` must be byte-for-byte identical to the committed ones; on a mismatch the script prints the exact claim path and both values.
4. **Manuscript audit.** `build_gate.py` reads every number out of `manuscript.tex` and requires each to be reproducible from `claims.json`. The abstract’s headline figures and every row of both tables are additionally anchored to their source, so a number swapped for a different valid one fails and is named. Exemptions — years, structural cuts, figures inside quoted 10-K text, and series identifiers — are listed explicitly in the source.
5. **Length.** The manuscript is checked against the journal’s word limit.

Determinism. The analysis has no stochastic component: no sampling, no bootstrap, no seed. `claims.json` and `Figure_1.png` are byte-identical across runs and across machines. `Figure_1.pdf` embeds a build timestamp and is therefore excluded from the byte comparison; its pixel content is verified through the PNG.

Environment. Python 3.11.5 (standard library only for every reported number), `matplotlib=3.8.0` (figure only), and, for the PDF only, `TeX Live 2024 / pdfTeX 1.40.26` with `elsarticle`, `siunitx`, `tabularx`, `booktabs`, `microtype`, `seqsplit`, `caption` and `lmodern`. Exact versions in `environment.txt`.

8 Data collection (not part of the reproduction)

`code/collect_*.py` document how the frozen inputs were originally collected: `collect_capex.py`, `collect_composition.py`, `collect_revenue.py` and `collect_alt_tags.py` pull the XBRL *frames* series; `collect_sic.py` pulls registrant classifications; `collect_ai_intensity.py` retrieves each firm's most recent 10-K and scores it against a dictionary fixed before any financing result was computed. This phase requires internet access and is intentionally excluded from the reproducible core: the *frames* interface returns the most recently filed value for each fact, so a later collection would pick up subsequent restatements. No API key is required at any point, and none is distributed. All reported results derive only from the frozen inputs.

9 Licensing and data availability

Code is released under the MIT Licence (see `LICENSE`); derived data, analysis outputs and text are released under CC-BY-4.0. The underlying raw data — SEC XBRL *frames*, SEC EDGAR filings and BLS series `WPUFD41312` — are U.S. public domain. The permanent citable archive is the Zenodo deposit at [\[\[DOI PENDING\]\]](#), which reproduces every reported number, table and figure offline, deterministically, and at zero cost.